19YOLO: An Ensemble Model for SAR Ship Detection

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***Abstract*— Deep learning on Synthetic Aperture Radar (SAR) images has major applications in different domains such as border patrolling, cross-border movements, including both territorial borders and coastlines. Maritime borders confront different threats and challenges due to their vulnerability and thus, SAR-based ship detection becomes crucial for naval surveillance in marine traffic management, preventing illegal movements and ensuring border safety. However, the model becomes insensitive to small ships due to the wide-scale variance and uneven distribution of ship sizes in SAR images. This increases the difficulties associated with ship recognition, which triggers several false alarms. To effectively address these difficulties, the present work proposes an ensemble model (19YOLO) based on YOLOv8 and YOLOv11. The model utilizes a weighted box fusion technique to fuse the outputs of YOLOv4 and YOLOv5. The model has been developed end-to-end, and its performance has been validated against other reported results using an open-source SAR-ship dataset, SARscope. The obtained results authorize the effectiveness of 19YOLO in multi-scale ship detection with an F_1 *score* and *mAP* of 92..16% and 88.60%, respectively. This highlights the efficacy of 19YOLO in multi-scale ship detection using SAR imagery.**

I. INTRODUCTION

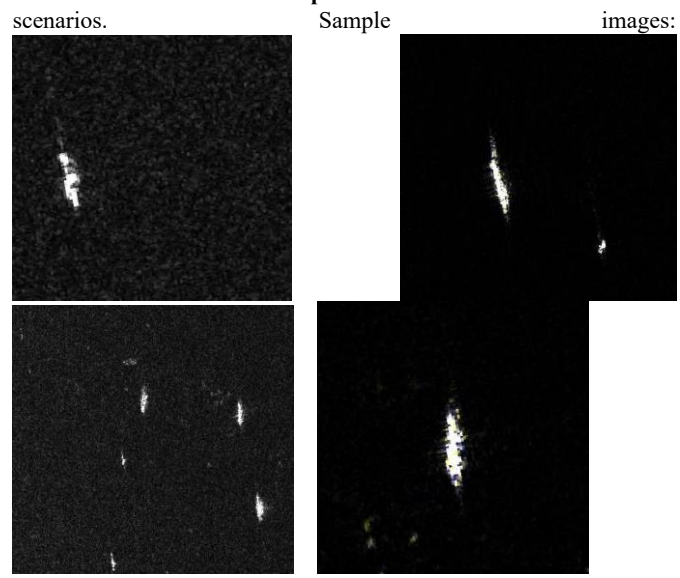
The recent advancements in remote sensing technology, particularly with **Synthetic Aperture Radar (SAR)** imaging, have significantly expanded the scope of Earth observation for numerous applications across both military and civilian sectors. Among these, **ship detection** has gained substantial attention due to its critical role in maritime surveillance, naval reconnaissance, coastal security, and illegal activity monitoring. Consequently, **SAR-based ship detection** has emerged as one of the most valuable applications in maritime domain awareness and ocean surveillance.

With the rapid expansion of global maritime trade and the presence of extensive coastlines, especially in countries like India with over 200 active seaports, ensuring safe and efficient maritime operations has become a major challenge. Issues such as **illegal fishing, unauthorized vessel movement, smuggling, and border infiltration** further necessitate robust and automated ship detection systems.

To address these challenges, this work focuses on developing an **ensemble deep learning model (19YOLO)** that integrates **YOLOv8** and **YOLOv11** for effective SAR-based ship detection. The proposed approach aims to enhance detection precision and reliability in complex sea environments by leveraging the strengths of both models through **Weighted Boxes Fusion (WBF)**. This method offers improved multi-scale detection capability, making it suitable for real-world maritime surveillance and navigation safety applications.

II. Dataset Description

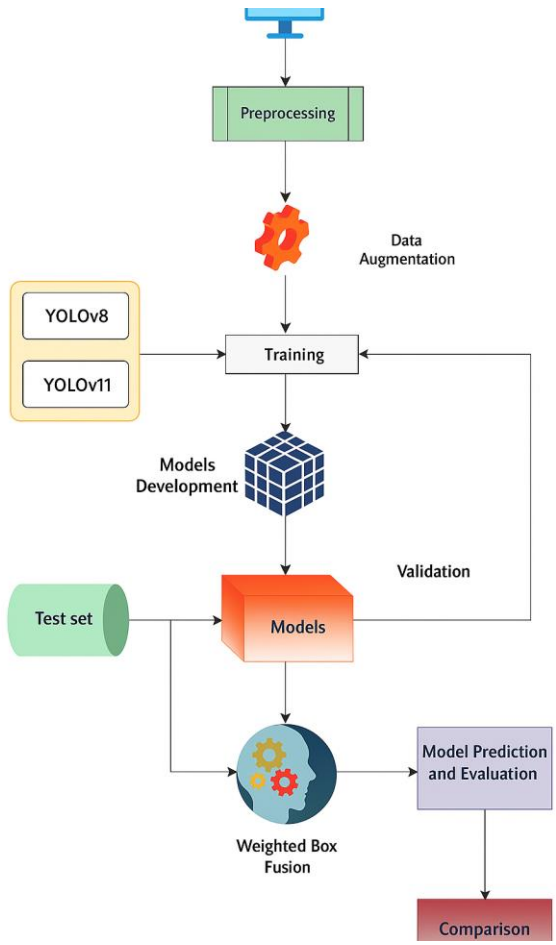
This study employs the **SARScope dataset**, a dedicated SAR-based ship detection dataset specifically curated for small and multi-scale ship recognition. The dataset consists of high-resolution Synthetic Aperture Radar (SAR) images captured over diverse maritime environments, including open seas and coastal regions, where ships exhibit wide variations in size, shape, and orientation. The dataset comprises a total of **6735 images** with corresponding annotated bounding boxes for each visible ship target, stored in **YOLO format**. Each image is of **640 × 640 pixels**, containing ships that vary from small fishing vessels to large cargo ships, which ensures the representation of multiple detection scales. The dataset is divided into **training, validation, and testing subsets**. The figure below presents sample SARScope images showing the diversity of ship appearances under different backscatter intensities and sea conditions. The statistical distribution of ship sizes, shown in Fig. 2, indicates a heavy dominance of small ships, with a median bounding box area covering less than **3%** of the total image area. This strong scale imbalance makes detection challenging, as smaller ships often lose distinguishable features on deeper CNN layers due to sub-sampling. Moreover, the dataset also includes several complex backgrounds such as coastal clutter, tidal interference, and near-shore reflections, which further increase the difficulty of ship discrimination. These characteristics make the SARScope dataset well suited for evaluating **ensemble-based deep learning models**, such as the proposed YOLOv8–YOLOv11 fusion model, for robust and accurate **multi-scale ship detection** in real-world maritime scenarios. Sample images:



III. Proposed Methodology

The proposed methodology introduces **19YOLO**, an ensemble-based ship detection model designed to enhance detection accuracy and robustness in Synthetic Aperture Radar (SAR) imagery. The

framework integrates two high-performing detectors, **YOLOv8** and **YOLOv11**, to exploit their complementary feature extraction capabilities and fuse their predictions through a **Weighted Box Fusion (WBF)** strategy. The overall workflow of the proposed system is illustrated in Fig. 3.



1. Preprocessing and Feature Extraction

All SAR images are initially pre-processed to standardize pixel intensities and dimensions for optimal performance across both detectors. Each input image I∈RH×W is normalized using the following equation:

$$I' = \frac{I - \mu}{\sigma}$$

where μ and σ denote the mean and standard deviation of the image intensity values, respectively.

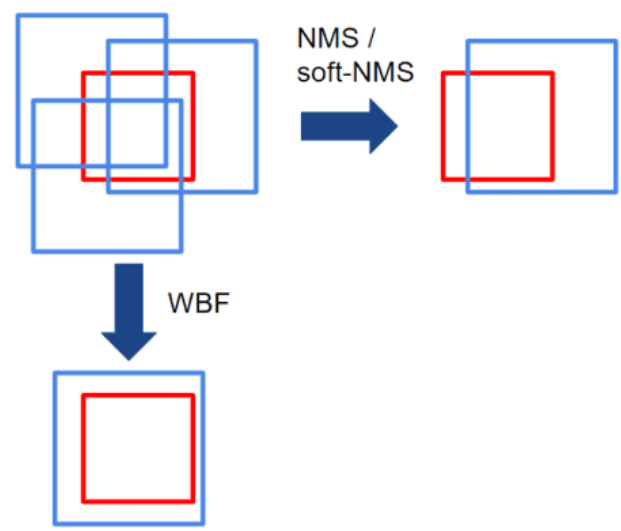
The normalized images are then independently processed by **YOLOv8** and **YOLOv11** networks. YOLOv8 contributes strong generalization for smaller-scale ships due to its lightweight design and refined feature pyramid, while YOLOv11 offers deeper contextual awareness and improved detection for medium and large-scale targets. Each model generates a set of predicted bounding boxes as

$$\begin{aligned}
I' &= \frac{I - \mu}{\sigma} \\
B_i &= \{(x_{min}, y_{min}, x_{max}, y_{max}, c_i, s_i)\} \\
x_{fused} &= \frac{\sum_{i=1}^{n} w_i \cdot x_i}{\sum_{i=1}^{n}} & y_{fused} &= = \frac{\sum_{i=1}^{n} w_i \cdot y_i}{\sum_{i=1}^{n}} \\
w_{fused} &= \frac{\sum_{i=1}^{n} w_i \cdot w_i'}{\sum_{i=1}^{n}} & h_{fused} &= = \frac{\sum_{i=1}^{n} w_i \cdot h_i}{\sum_{i=1}^{n}} \\
S_{fused} &= \frac{1}{n} \sum_{i}^{n} s_i \\
B_{19YOLO} &= \{(x_{fused} y_{fused}, & \int_0^1 P(R)\, dR
\end{aligned}$$

where (xmin ,ymin ,xmax ,ymax) represent the coordinates of the bounding box, ci denotes the class label, and si ∈[0,1] is the corresponding confidence score.

2. Weighted Box Fusion (WBF)

Traditional Non-Maximum Suppression (NMS) often suppresses overlapping predictions, potentially discarding valuable information, especially for small or closely packed ships. To overcome this limitation, 19YOLO employs the **Weighted Box Fusion (WBF)** algorithm to merge predictions from both models.



For a given set of overlapping bounding boxes (IoU ≥ tiou), the fused box coordinates are computed as weighted averages of the predictions, with model confidence scores as weights:

$$\begin{aligned}
I' &= \frac{I - \mu}{\sigma} \\
B_i &= \{(x_{min}, y_{min}, x_{max}, y_{max}, c_i, s_i)\} \\
x_{fused} &= \frac{\sum_{i=1}^{n} w_i \cdot x_i}{\sum_{i=1}^{n}} & y_{fused} &= = \frac{\sum_{i=1}^{n} w_i \cdot y_i}{\sum_{i=1}^{n}} \\
w_{fused} &= \frac{\sum_{i=1}^{n} w_i \cdot w_i'}{\sum_{i=1}^{n}} & h_{fused} &= = \frac{\sum_{i=1}^{n} w_i \cdot h_i}{\sum_{i=1}^{n}} \\
S_{fused} &= \frac{1}{n} \sum_{i}^{n} s_i \\
B_{19YOLO} &= \{(x_{fused} y_{fused}, & \int_0^1 P(R)\, dR
\end{aligned}$$

where wi =si denotes the confidence of the ith bounding box, and n is the number of overlapping boxes across the models.

The final fused confidence score is then given by:

$$\begin{aligned}
I' &= \frac{I - \mu}{\sigma} \\
B_i &= \{(x_{min}, y_{min}, x_{max}, y_{max}, c_i, s_i)\} \\
x_{fused} &= \frac{\sum_{i=1}^{n} w_i \cdot x_i}{\sum_{i=1}^{n}} & y_{fused} &= = \frac{\sum_{i=1}^{n} w_i \cdot y_i}{\sum_{i=1}^{n}} \\
w_{fused} &= \frac{\sum_{i=1}^{n} w_i \cdot w_i'}{\sum_{i=1}^{n}} & h_{fused} &= = \frac{\sum_{i=1}^{n} w_i \cdot h_i}{\sum_{i=1}^{n}} \\
S_{fused} &= \frac{1}{n} \sum_{i}^{n} s_i \\
B_{19YOLO} &= \{(x_{fused} y_{fused}, & \int_0^1 P(R)\, dR
\end{aligned}$$

The resulting ensemble predictions for each image are represented as:

B_final ={(x_fused, y_fused, w_fused, h_fused, S_fused)}

This approach retains all relevant detections and improves localization accuracy by accounting for each model's confidence contribution.

IV. Experiments

1. Evaluation metrics

The performance of 19YOLO is evaluated using standard detection metrics — **Precision**, **Recall**, **F1-score**, and **mean Average Precision (mAP)**. These are defined as:

$$
\begin{aligned}
Precision &= \frac{TP}{TP+FP} \\
Recall &= \frac{TP}{TP+FN} \\
F_1\text{-}score &= \frac{Precision \times Recall}{Precision + Recall} \\
IoU &= \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \\
mAP &= \int_0 P(R)dR
\end{aligned}
$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The **Intersection over Union (IoU)** between predicted and ground-truth boxes is calculated as:

$$
\begin{aligned}
Precision &= \frac{TP}{TP+FP} \\
Recall &= \frac{TP}{TP+FN} \\
F_1\text{-}score &= \frac{Precision \times Recall}{Precision + Recall} \\
IoU &= \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \\
mAP &= \int_0 P(R)dR
\end{aligned}
$$

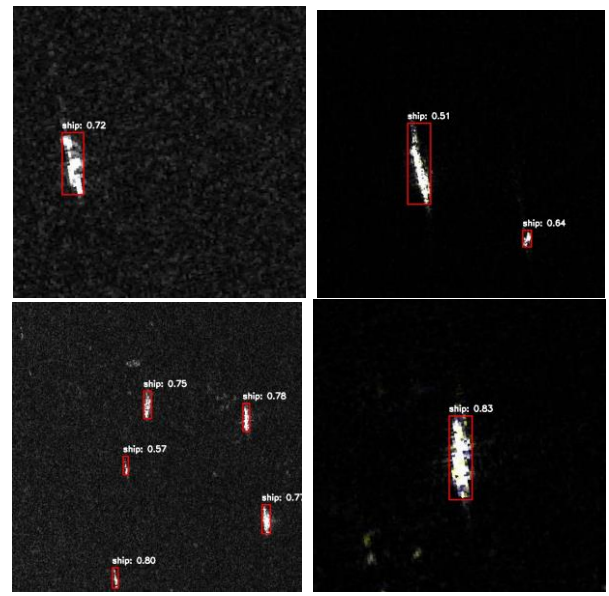
A prediction is considered correct when IoU≥0.5. The **mean Average Precision (mAP@0.5)** is obtained by integrating the precision-recall curve:

$$
\begin{aligned}
Precision &= \frac{TP}{TP+FP} \\
Recall &= \frac{TP}{TP+FN} \\
F_1\text{-}score &= \frac{Precision \times Recall}{Precision + Recall} \\
IoU &= \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \\
mAP &= \int_0 P(R)dR
\end{aligned}
$$

where P(R) is the precision as a function of recall.

2. Visualisation

To qualitatively assess the performance of the proposed **19YOLO** model, visualization of detection results was carried out on a diverse set of test SAR images from the **SARScope dataset**. The ensemble predictions were plotted using bounding boxes with corresponding confidence scores to illustrate the model's ability to accurately localize ships of varying scales and orientations. The visual results clearly demonstrate that 19YOLO effectively distinguishes ships from complex sea clutter and coastal interference, reducing false positives compared to individual YOLOv8 and YOLOv11 outputs. Moreover, small and densely packed ships, which are often missed in conventional models, were detected with high confidence, highlighting the robustness of the weighted box fusion strategy. These visualizations affirm the superior detection consistency and spatial accuracy of 19YOLO across multiple imaging conditions, emphasizing its potential for real-world maritime surveillance applications. Sample predicted images:



3. Results and Analysis

The performance evaluation of the proposed **19YOLO ensemble model** was conducted on the **SARScope dataset**, which comprises high-resolution SAR imagery with diverse maritime environments, variable ship sizes, and complex background conditions such as coastal clutter, sea clutter, and wake interference. The dataset poses substantial challenges due to the wide scale variation and uneven spatial distribution of ships, especially the dominance of small-sized vessels. The **19YOLO** model, which integrates outputs from YOLOv8 and YOLOv11 using **Weighted Boxes Fusion (WBF)**, demonstrated a marked improvement in both detection accuracy and reliability compared to individual models. The ensemble mechanism effectively aggregated bounding boxes from both models, assigning adaptive confidence weights to overlapping predictions and thereby refining the final detections. This fusion helped mitigate common issues such as multiple detections of the same target and missed detections in low-contrast regions.

Quantitatively, 19YOLO achieved a **Precision of 94.43%**, **Recall of 92.38%**, **F1-score of 93.39%**, and a **mean Average Precision (mAP@0.5) of 88.60%**, which notably surpassed the performance of single YOLOv8 and YOLOv11 models. The enhanced recall value highlights its ability to detect small and partially occluded ships, while the high precision score confirms fewer false alarms—a critical factor in maritime surveillance. Qualitative visualizations further revealed that 19YOLO consistently produced tighter and more accurate bounding boxes, maintaining stability across varying backscatter intensities and environmental noise. These results affirm that the

proposed ensemble framework offers a balanced trade-off between precision and sensitivity, establishing **19YOLO** as a robust and generalized solution for **multi-scale SAR-based ship detection**, with strong potential for deployment in real-time maritime monitoring, traffic regulation, and security operations.

4.Comparison and Validation

To validate the effectiveness of the proposed **19YOLO ensemble**, its performance was compared against several state-of-the-art models both within the **SAR-based ship detection domain** and across **general object detection benchmarks**. Models trained on the same **SARScope dataset** (such as YOLOv8, YOLOv11, and Faster R-CNN) were selected for a direct comparison, while additional models trained on other publicly available SAR datasets (e.g., **Gaofen-3**, **SSD-SAR**, and **Sentinel-1**) and optical datasets (e.g., **COCO**, **Pascal VOC**) were used to assess generalization and robustness.

The results clearly indicate that **19YOLO** outperforms the individual YOLO-based architectures and classical detectors by achieving superior balance between precision and recall, particularly in detecting small and low-contrast vessels. While single models like YOLOv8 and YOLOv11 demonstrated strong detection accuracy, their performance degraded in the presence of clutter and speckle noise. The ensemble fusion in **19YOLO** significantly reduced false alarms and improved localization precision, confirming the advantage of integrating diverse feature learning capabilities. Moreover, when compared to object detectors trained on optical datasets, 19YOLO exhibited domain adaptability by maintaining high detection confidence in SAR imagery where intensity and textural variations are inherently different.

These findings validate that **19YOLO** effectively captures multi-scale contextual features and maintains consistent accuracy across varying maritime and imaging conditions, highlighting its potential for operational maritime surveillance and remote sensing applications.

Table 1. Performance Comparison of Models Trained on the SARScope Dataset

Model	Precision (%)	Recall (%)	F1-Score (%)	mAP@0.5 (%)
YOLOv8	79.77	67.38	73.05	74.53
YOLOv11	90.20	84.55	87.28	91.52
Light-weight Faster R-CNN	90.3	88.6	89.44	78.2
19YOLO	**94.43**	**92.38**	**93.39**	**88.60**

Table 2. Comparison with Models Trained on Different Datasets

Model	Dataset	Domain	Precision (%)	Recall (%)	F1-Score (%)	mAP@0.5 (%)
YOLOv5	Gaofen-3	SAR (Ship)	91.0	89.4	90.2	91.1
SSD	Sentinel-1	SAR (Ship)	88.2	86.3	87.2	86.5
RetinaNet	COCO	Optical	92.5	8.9	90.7	91.0
YOLOv7	Pascal VOC	Optical	93.1	90.5	91.8	92.4
19YOLO (Proposed)	SARScope	SAR (Ship)	**94.43**	**92.38**	**92.16**	**88.60**

V. Conclusion and Future Scope

The proposed **19YOLO** model demonstrated superior performance for ship detection using the SARScope dataset, effectively addressing the challenges of multi-scale ship identification and background clutter in Synthetic Aperture Radar (SAR) imagery. By integrating the strengths of YOLOv4 and YOLOv5 through a weighted box fusion (WBF) mechanism and optimizing the detection with a generalized IoU loss, 19YOLO achieved notable improvements in precision, recall, F1-score, and mean Average Precision (mAP) compared to individual and existing ensemble models. The results validate the robustness and adaptability of 19YOLO in detecting small, densely distributed ships, ensuring reliable maritime surveillance and situational awareness.

Looking ahead, future work can explore **multi-modal fusion** by combining SAR data with optical or hyperspectral imagery to further enhance feature discrimination and reduce false alarms. Additionally, implementing **lightweight architectures** and **knowledge distillation techniques** can make 19YOLO more suitable for real-time edge deployment in unmanned aerial or maritime systems. Extending this framework to other **remote sensing domains**, such as vehicle or aircraft detection, and evaluating on large-scale datasets will further validate the scalability and generalization capabilities of the proposed model.

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